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FLEA NEWS is a biannual newsletter devoted to matters involving insects belonging to the order Siphonaptera (fleas) and related subjects. It is compiled and distributed free of charge by Robert E. Lewis (relewis@iastate.edu) with the support of the Department of Entomology at Iowa State University, Ames, IA, and a grant in aid from **Sandoz Animal Health**, based in Des Plaines, IL. It is mainly bibliographic in nature. Many of the sources are abstracting journals and title pages and not all citations have been checked for completeness or accuracy. Additional information will be provided upon written or e-mail request. Further, recipients are urged to contribute items of interest to the profession for inclusion herein.

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MISCELLANEA

Shortly after sending out Flea News 53 I received a letter from Mr. F.G.A.M. Smit, past curator of the Rothschild Collection of fleas in the British Museum (Natural History). It contained excerpts from a letter from István Szabó, a past member of the Hungarian National Museum and student of Siphonaptera. "Pista" celebrated his 84th birthday on June 13th and wrote to describe some of the honors bestowed upon him during 1996. Since some of the recipients of this newsletter are familiar with his work, Frans felt that part of the account would be of interest. Following are paraphrased quotes from part of the letter:

"1996 was a very rewarding year for me. First I was awarded the **Pro-Natura** medal and certificate for my "outstanding nature conservation activities". This award is presented annually to three or four recipients. I have been working extensively on the conservation of the Hungarian fauna, specifically the mammals. I proposed a conservation law protecting all species of bats, shrews, the badger and the mole, as well as a few species of field mice. Such proposals usually require a lengthy explanation and justification, but mine was accepted without change.

"Each year, at the spring opening of the woodcock hunting season, I deliver a speech as part of duties I assumed a few years ago. This spring, to my surprise, the President of the National Hunting Association awarded me the **Gold Hubertus Cross** for my "outstanding hunting ethics". This award has only been presented to 18 persons throughout its history, and receiving it gave me great satisfaction.

"Finally, in September, came the most rewarding presentation of all; an **Hungarian Knighthood**. This is the only award given exclusively for military service. It was awarded for my service in the military fifty years ago during WW II and for my two years on the front lines. During that period I received many medals; the **Signum**

Laudis, Cross of Fire ...etc. Since sworn to the knighthood, I can now use the term *Vitez* (Knight) with my name and be called *Nemzetes ur* (Mr. Nationalist), which is a kind of Sir in Hungarian. Unfortunately this title had a great price and I am fortunate to be alive to receive it. Not only did I look death in the eye on a daily basis, but spent two years in a prison camp in Siberia, which was worse than any battles. Minus 20 to 30 degrees C in winter and plus 40 degrees C in the summer. We cut wood and worked in a stone quarrey. All of my extremities were frozen and it took years for me to recover. We wore little clothing and were always hungry. When I finally came home I weighed less than 100 pounds, but I was always lucky, and God took good care of me."

I'm sure we all wish Pista many happy returns on his 84th birthday, and many more to come!

Following is an announcement of the **Third International Symposium on Fleas**.

We are pleased to inform you that the Third International Symposium on Fleas will be held in Baicheng, Jilin Province, sponsored by the Chinese Centre for Treatment and Prophylaxis of Plague and Brucellosis, in accordance with the resolution of the Second International Symposium on Fleas held at Stavropol [, USSR] in 1994. You are cordually invited to attend the symposium, which will provide excellent opportunities for scientists and specialists to exchange views and experiences.

The program of the symposium will include, but not be limited to, presentations on the systematics, distribution and ecology of fleas.

The dates are September 25-26, 1997.

Financial arrangements do not include transportation from your country of origin to Beijing-Changchun, but round-trip travel expenses between Changchun and Baicheng will be provided.

If you intend to attend the symposium, please send the title and abstract (in English) to us as soon as possible. Otherwise, if you cannot be present, we would very much like to include your presentation in the Proceedings with your approval. Submission deadline is 30 June 1997.

For further information please contact Mr. MA Huan-wei at:

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The following report on the status of plague in mainland China was published in the **Vector Ecology Newsletter** 28(1):4-5 (1997) from a report submitted by Mr. XU Rong-man, Institute of Microbiology and Epidemiology, 20 Dongdajie, Fengtai, Beijing 100071, CHINA. It is reprinted with permission from SOVE:

"Through more than forty years of effort, rodent control efforts have succeeded and human plague has decreased; only 13 cases in 1993, zero cases in 1994, 8 cases in 1995, and 21 cases in 1996. But we must be on the alert because there are ten geographical foci of plague in China. The following illustrates the situation ...

1. Plague focus [in] the house rodent, *Rattus flavipectus*. This species is found in southern Yunnan and seaboard of Zenjiang, Fujian, Taiwan, Guangdong and Guangxi in southern China, covering over 20,000 sq. km., including 56 counties. The main vectors are *Xenopsylla cheopis* and *Leptopsylla segnis*. Other infected flea [species] are [*Ceratophyllus*] (*Monopsyllus*) *anisus* and *Nosopsyllus nicanus*. Other infected [hosts] are *Rattus norvegicus*, *Mus musculus* and *Suncus murinus*. Since 1953 no human plague has occurred [in this focus] except in south Yunnan.

2. Plague focus [in] *Eothenomys miletus* in the mountains. Its distribution is in northwestern Yunnan, covering 600 sq. km. The main vectors are *Ctenophthalmus quadratus* [s], and to a lesser degree, *Neopsylla specialis*. The main reservoir host is *Eothenomys miletus*. Other infected animals are *Apodemus chevrieri*, *Apodemus speciosus* and *Rattus nitidus*. This wild rodent plague focus was identified in 1975. Epizootic plague has been found many times in the past 20 years, but no human plague has been reported.

3. Plague focus [in] *Marmota himalayana*. [This host] is found mainly in Tibet and Qinghai, south to the Himalaya Mountains, north to the Qilian Mountains, west to the Kunlun Mountains in Xinjiang, and east to southern Gansu, covering nearly 1,000,000 sq. km. [and] including 54 counties. The main vectors are *Callopsylla dolobris* and *Oropsylla silantiewi*. Other infected fleas and [hosts] are *Rhadinopsylla li* [and] *Pulex irritans* and *Ochotona daurica annectans*, *Ochotona curzoniae*, *Lepus oiostolus*, *Vulpes ferrilata*, *Procapra picticauda*, *Mus musculus*, *Cricetulus migratoriu[s]*, *Microtus oeconomus* and *Pitymys leucurus*. This is a stable epizootic focus and is active from April to September. {It} is the most important focus in China and produces most of the human cases.
4. Plague focus [in] *Marmota caudata*. Its distribution is in southwestern Xinjiang and is a part of the Pamir Plateau plague focus in Middle Asia, covering 600 sq. km., including two counties. The main vectors are *Oropsylla silantiewi* and *Rhadinopsylla li*. One other infected flea species and [mammal host] is *Citellophilus lebedewi priceps* and *Pitymys juldaschi*. This focus was identified in 1956 and is active from May to August. There has been no human plague recorded in the epizootic plague focus.
5. Plague focus [in] *Marmota baibacina* and *Spermophilus undulatus*. Its distribution is in the Tianshan Mountains of Xinjiang, encompassing 7,000 sq. km., including 10 counties. This focus extends in[to] Kazakhstan and Kirgizstan. The main vector is *Oropsylla silantiewi*. Other infected flea [species and hosts] are *Callopsylla dolabris*, *Citellophilus tesquorum altaicus* and *Clethrionomys glareolus*. Here epizootic plague occurs from May to September. Since 1973 no human plague cases have been recorded.
6. Plague focus [in] *Spermophilus alaschanicus* in Gansu-Ningxia. Its distribution is in eastern Gansu and southern Ningxia, northern China, encompassing 3,000 sq. km., including five counties. The main vector is *Citellophilus tesquorum mongolicus*. Other infected flea [species] are *Neopsylla abagaitui*, *Frontopsylla elat[a]* and *Ophthalmopsylla praepecta*. Other infected host [species] are *Myospalax fontanieri*, *Meriones meridianus*, *Cricetulus triton*, *Al[l]actaga sibirica* and *Ochotona daurica*. Epizootic plague occurs [from] April to October. Since 1978 no human plague cases have been recorded.
7. Plague focus [in] *Meriones unguiculatus* in [the] Inner Mongolian plateau. Its distribution is mainly in Inner Mongolia, and the nearby three counties [=Provinces] of Ningxia, Shaanxi and Hebei, encompassing 100,000 sq. km., including 28 counties. The main vectors are *Nosopsyllus laeviceps* and *Xenopsylla conformis*. Other infected flea [species] and [hosts] are *Neopsylla pleskei*, *Citellophilus tesquorum mongolicus*, *Paradoxopsyllus kalabukovi*, *Rhadinopsylla insolita* [and] *Rhadinopsylla tenella*, and *Spermophilus dauricus*, *Spermophilus erythrogenys*, *Meriones meridianus*, *Dipus sagitta* and *Mus musculus*. Epizotic plague appears [from] April to November. There was a large epidemic in animals in 1970-1971. Since 1973 no cases of human plague have been reported.
8. Plague focus [in] *Spermophilus dauricus* in the plains of [the] Songhuajiang - Liaohe Rivers. Its distribution is in Inner Mongolia, Liaoning, Jilin and Heilongjiang provinces, covering 120,000 sq. km., including 42 counties. The main vector is *Citellophilus sungaris*. Other flea and host species are *Neopsylla bidentatiformis* and *Xenopsylla cheopis* and *Rattus norvegicus* and *Mus musculus*. There are small epidemics in animals every three to five years and larger epidemics every 10-15 years. There have been no human cases of plague since 1959.
9. Plague focus [in] *Microtus brandti* in [the] Xilin Gol Plateau. Its distribution is in the Xilin Gol Banner [?] of northern Inner Mongolia, with 60,000 sq. km. The main vectors are *Amphipsylla primaris* and *Neopsylla pleskei*. Other infected flea and host species are *Frontopsylla luculenta*, *Neopsylla bidentatiformis*, *Citellophilus tesquorum mongolicus*, *Nosopsyllus laeviceps* and *Meriones unguiculatus*, *Spermophilus dauricus*, *Ochotona daurica*, *Allactaga sibirica* and *Mus musculus*. Epizootic plague occurs from April to November. It seems that there is an epidemic every five years. No plague in humans has occurred historically.
10. Plague focus [in] *Marmota bobac sibirica* in [the] Hulum Buir Plateau. Its distribution is in northeastern Inner Mongolia, with 40,000 sq. km. This epizootic focus is part of the *Marmota bobac sibirica* focus in Russia and Mongolia. The main vector is *Oropsylla silantiewi*. Since 1926 the plague bacillus has not been isolated from marmots [or their] fleas, therefore this plague focus has been eradicated. It may be due to hunting marmots for a long period of time.

The following was printed in the Technical Information Bulletin of the Armed Forces Pest Management Board, March-April issue, and originally appeared in volume 10(2) of Infectious Disease News.

over the decade 1984 to 1994, a total of 18,739 cases of human plague were reported to the World Health Organization from 20 countries, according to the Centers for Disease Control and Prevention. In the United States, 341 cases of human plague were reported between 1970 and 1995; 80% of these occurred in New Mexico, Arizona and Colorado, usually during the summer. In response to these cases, the Advisory Committee on Immunization Practices (ACIP) last year updated its recommendations concerning plague vaccination, antibiotic treatment and public health measures.

Killed *Yersinia pestis* have been standard in plague vaccines since 1896; however, only one vaccine, a formalin-inactivated preparation, is currently licensed for use in the United States. Live *Y. pestis* vaccines comprising presumably avirulent strains have been developed, but the ACIP report states that they are not commercially available because their safety and efficacy have not been adequately tested. Because of the low incidence of plague in this country, even the efficacy of the inactivated vaccine has not been measured in controlled studies on humans. Moreover, researchers have yet to determine whether vaccination protects against infectious droplets or aerosols generated in laboratories or against exposure to infectious respiratory droplets from patients with pneumonic plague. Data concerning the development of protective antibodies in vaccinated humans are also limited.

The ACIP report clearly states that only persons at high risk for exposure should be vaccinated against plague. Those at high risk include laboratory personnel who routinely handle viable *Y. pestis* cultures and those who have regular contact with wild rodents or their fleas in areas where plague is enzootic or epizootic. If vaccinated individuals have been exposed to plague or if their risk of exposure is exceptionally high (as among health workers caring for patients with pneumonic plague), antibiotic prophylaxis should be considered as a supplement to vaccination. Vaccination is not intended to control plague. Routine vaccination is not indicated for those living in plague-enzootic areas, nor is it necessary for hospital or other medical personnel in such areas, or for travelers to countries where cases have been reported. Primary plague vaccination is available only for people aged 18 to 61; safety and immunogenicity for people younger than 18 and effects on developing fetuses are unknown.

The ACIP report lists seven steps for reducing the risk of acquiring plague: eliminate sources of food and shelter for rodents near homes; modify homes to prevent rodent access; avoid rodent nests and burrows; avoid direct contact with sick or dead animals of any kind; treat domestic dogs and cats weekly with appropriate insecticides; handle sick cats with extreme caution; and use insect repellents containing DEET on skin as well as repellents or appropriate insecticidal sprays on clothing when engaged in recreational activities

NOMENCLATRURAL TIDBITS

Those of you who are long-time recipients of Flea News will be familiar with at least some of the following, but with the 21st Century bearing down on us and the effect it will have on our computerization systems, a brief history of the project seems appropriate.

In 1978, Mr. Frans Smit, then Curator of Siphonaptera at the British Museum (Natural History) and Alynn Wright, Smit's assistant, sent out two major nomenclatural compilations to recipients of Flea News. The first of these, 1978a, was "*A list of code numbers of species and subspecies of Siphonaptera*" (49 pp.). The second, 1978b, was "*A catalogue of primary type-specimens of Siphonaptera in the British Museum (Natural History)*" (71 pp.). While the latter was a stand-alone document which is as useful today as it was in 1978, the former has required continuous updating and modification.

Anticipating the development of computerized data storage, Smits numbering system was based upon the chronology of publication, both by years and within the year. Since our nomenclatural system is based on Linnaeus' 1758 edition of *Systema Naturae* the first entries are for the two named species contained therein; *Pulex irritans* L., 1758 and *Pulex penetrans* L., 1758. The former was assigned the number 75801 since it appears first, the latter, 75802 since it appears last. Based on this system, each name of a species or subspecies was assigned its own unique number until Mr. Smit retired in 1980. Since then I have attempted to continue to upgrade the list annually, though assignment of the numbers has lagged behind the actual year of description in order to be relatively certain that all of the names for a given year had been retrieved. Fewer and fewer new taxa have been described in recent years and the last year for which numbers were assigned was 1992. Since there are so few active systematists in the order now the urgency to keep the file current has diminished, although I have continued to maintain it in my personal data base

Now a relational data base program has been developed by the Commonwealth Scientific and Industrial Research Organization of Australia. Appropriately called [PLATYPUS](#) after one of the two unique, egg laying types of mammals of the area, this program promises to revolutionize the construction of faunal data bases. I have only recently purchased it, and whether it is a characteristic of the software or of my generation, I have yet to master its intricacies to the extent that I can

describe it accurately. However, it appears at this point to be ideal for the taxonomist, for which it was originally designed, and I hope to be able to report on it in detail in the December number of this newsletter. The whole system is based on a central "checklist" which is really an alphabetical inventory of taxa at the family level or below. However, there are two "levels" above family which fit it admirably to the ordinal and subordinal or superfamilial categories, thus creating a hierarchy, in descending order of magnitude, from order/suborder/superfamily; through family, subfamily, tribe, genus, subgenus, species, and subspecies, with all of the attendant information about authors, date, citations and synonymy, plus almost unlimited nomenclatural, ecological, and distributional information, as well as keyword associations at various levels. I hope to be able to discuss the merits of this software in the next issue.

SIPHONAPTERA LITERATURE

The amount of literature that has appeared since mailing Flea News 53 has been remarkably small, especially that dealing with systematics and nomenclature. As usual, though it may not be obvious from the titles, citations included here pertain to fleas and the zoonoses associated with them. No particular effort has been made to search the medical and veterinary literature and the emphasis here is on the taxonomy, systematics and general biology of members of the order.

1994 (List 7)

LIN Ting-hsian & CHUNG Chao-lin. A new subspecies of the genus *Stivalius* (Siphonaptera: Pygiopsyllidae) from Taiwan. Annual of the Taiwan Museum 37: 23-29. (*Stivalius aporus yenpinensis*).

1995 (List 5)

Bardt, D. Untersuchungen zur Wirkung des Insect Growth Regulator Pyriproxyfen auf den Katzenfloh *Ctenocephalides felis* (Bouché) und Entwicklung einer Formulierung zur Flohbekämpfung und der Katz. Inaugural-Dissertation, Fachbereich Veterinärmedizin, Freie Universität, Berlin, Germany. 115 pp.

CHEN Jing-long & WANG Dun-qing. Preliminary studies on restriction enzymatic analysis of genomic DNA in three species of domestic rat fleas in Fujian. Endemic Diseases Bulletin 19(3): 8-11.

Franz, M. & M.C. Cadiergues. Use of lufenuron in the control of canine infections with *Ctenocephalides felis*. Medicina Veterinaria 12(10): 598-602.

Imai, S., M. Takeda, T. Uchino, M. Makano & Y. Kotake. Species distribution of fleas infesting dogs and cats in Japan. Journal of the Japan Veterinary Medical Association 48(10): 775-778.

Radulovic, S., J.A. Higgins, D.C. Jaworski, G.A. Dasch & A.F. Azad. Isolation, cultivation and partial characterization of the ELB agent associated with cat fleas. Infection and Immunity 63(12): 4826-4829.

Sapegina, V.F., N.L. Gershkovich, Yu.V. Drozdova, I.V. Luk'yanov & Yu.S. Ravkin. The flea fauna (Siphonaptera) of the northeastern Altai. Entomologicheskoe Obozrenie 74(3): 582-588.

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ZHANG Rong-guang, WU De-qiang & DENG Kai-ze. Ecological studies on parasitic fleas of *Marmota himalayana* in Gansu Province. Endemic Disease Bulletin 10(3): 52-55.

1996 (List 3)

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Braack, L.E., I.G. Horak, L.C. Jordaan, J. Segerman & J.P. Louw. The comparative host status of red veld rats (*Aethomys chrysophilus*) and bushveld gerbils (*Tatera leucogaster*) for epifaunal arthropods in the south Kruger National Park, South Africa. Onderstepoort Journal of Veterinary Research 63(2): 149-158.

Christe, P., H. Richner & A. Oppliger. Of great tits and fleas - sleep baby sleep. Animal Behaviour 52(6): 1087-1092.

Darskaya, N.F. & V.M. Malygin. On the fleas of mammals in the basin of the Ucayali river basin (Peruvian Amazonia) Parazitologia 30(2): 187-190.

Dryden, M.W. & B.L. Reid. Insecticide susceptibility of cat flea (Siphonaptera: Pulicidae) pupae. Journal of Economic Entomology 89(2): 421-427.

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Hoogland, J.L. *Cynomys ludovicianus*. Mammalian Species 535: 1-10.

Jackson, J.E., L.C. Branch & D. Villarreal. *Lagostomus maximus*. Mammalian Species 543: 1-6.

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Medvedev, S.G. Fleas of the family Ischnopsyllidae (Siphonaptera) of the fauna of Russia and adjacent countries Entomologicheskoe Obozrenie 75(2): 438-454.

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Rydell, J. & W. Bogdanowicz. *Barbastella barbastella*. Mammalian Species 557: 1-8.

Schlitter, D.A. & M.B. Qumsiyeh. *Rhinopoma microphyllum*. Mammalian Species 542: 1-5.

Sleeman, D.P., P. Smiddy & P. Moore. The fleas of Irish terrestrial mammals: a review. The Irish Naturalist's Journal 25(7): 237-248.

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Murray, J.L. & G.L. Gardner. *Leopardus pardalus*. Mammalian Species 548: 1-10.

Whiting, M.F., J.C. Carpenter, Q.D. Wheeler & W.C. Wheeler. The Strepsiptera problem: Phylogeny of the holometabolous insect orders inferred from 18S and 28S ribosomal DNA sequences and morphology. *Systematic Biology* 46(1): 1-68.